TUTORIAL

Tutorial: structured models — more practice

Seven further problems: two on the Williams and LAB models, five on the cybernetic model. Each has a solution and an 'explained' slide.

Williams and LAB models

5 **Williams: from data to wash-out**

Fit k_2, k_3 from active-fraction data; D_{max} and s_{min}

Builds on Problem 1

6 **LAB: best productivity**

Biomass and productivity vs D for a 2 g L^{-1} feed

Builds on Problem 2

Cybernetic model

7	**Fully induced cells, three sugars**	u, w with both enzymes present; switch point; a third sugar
8	**Glucose pulse into a xylose culture**	Why the better sugar is ignored for hours; role of k_{con}
9	**The whole diauxic batch**	Glucose phase, switch time, lag and xylose phase by hand
10	**Chemostat on one sugar**	Closed-form steady state, D_{max}, comparison with Monod
11	**Mixed feed in a chemostat**	Simultaneous use at low D, sequential use at high D

How to use: as before, hide the solution and explained slides (right-click → Hide Slide) and reveal them after students have worked each problem. A calculator is enough for every problem; 8(d), 9 and 11 also compare hand estimates with exact answers. Problems 7–11 use the Problem 4 parameters unless stated.

Parameter values are illustrative, chosen to give realistic magnitudes; they are not fitted values from the original papers. Answers were checked by numerical integration.

Problem 5 — Williams model: from data to wash-out

Same Williams culture as Problem 1, but k_2 and k_3 must first be read from measurements

Given

Symbol	Value	Units
γ_{11}	0.50	g X_A (g S)$^{-1}$
γ_{22}	0.90	g X_G (g X_A)$^{-1}$
k_1	2.0	g S (g DW h)$^{-1}$
K_s	0.020	g L^{-1}

Measured active fraction at steady state

D (h^{-1})	0.05	0.20	0.40	0.60
X_A	0.056	0.111	0.185	0.259

Model

$$v_1 = k_1 \frac{s}{s + K_s}, \quad v_2 = k_2 X_A X_G, \quad v_3 = k_3 X_G$$

$$\mu = \gamma_{11} v_1 - (1 - \gamma_{22}) v_2, \qquad X_A + X_G = 1$$

$$X_A = \frac{D + k_3}{\gamma_{22} k_2} \quad \text{(Problem 1a)}$$

Tasks

(a) Plot X_A against D. From the slope and intercept of $X_A = (D + k_3)/(\gamma_{22}k_2)$, estimate k_2 and k_3.

(b) At wash-out glucose is in excess, so f → 1. Show that D_{max} obeys $D = \gamma_{11}k_1 - (1 - \gamma_{22})k_2 X_A(1 - X_A)$ with X_A from (a). Solve it by iteration, starting from D = 1.0 h^{-1}.

(c) Find the residual glucose s at D = 0.8 and 0.9 h^{-1}. What does this mean for running close to D_{max}?

(d) Set D = 0 to find s_{min}, the lowest glucose level at which the culture can be maintained. Compare with the LAB value 12.3 mg L^{-1} (Problem 2d).

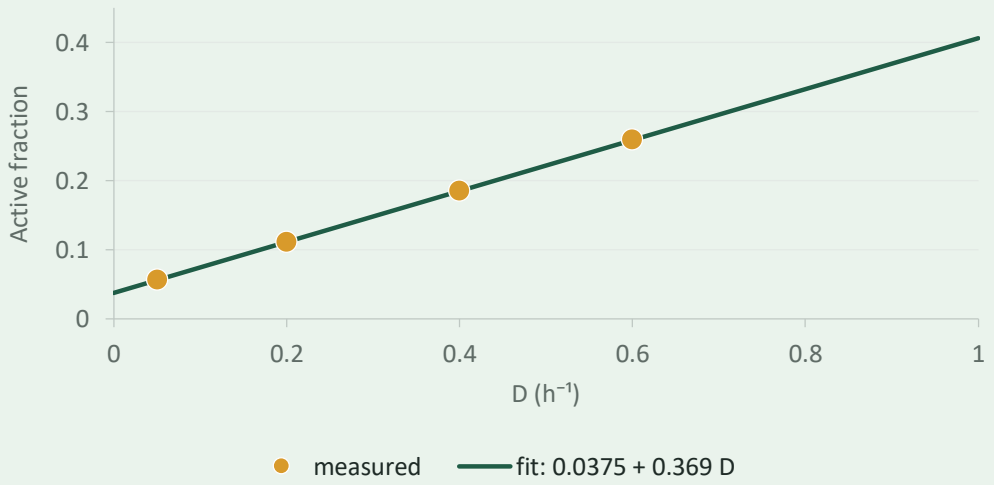
Check your work: slope ≈ 0.37 h, so $\gamma_{22}k_2 \approx 2.7$ h^{-1}; the iteration settles at $D_{max} \approx 0.93$ h^{-1}.

Builds on Problem 1 (VNL eqs. 7.32a–c, Ex. 7.5). 'Measured' X_A values generated with $k_2 = 3.0$ h^{-1} and $k_3 = 0.10$ h^{-1}, then rounded.

Solution 5 — Williams model from data

The straight line gives the parameters; two iterations give the wash-out rate

(a) Fit of X_A against D



Slope = (0.259 − 0.056)/(0.60 − 0.05) = 0.369 h ⇒ $\gamma_{22}k_2$ = **2.71 h^{-1}**, k_2 = 2.71/0.9 = **3.0 h^{-1}**

Intercept = 0.056 − 0.369 × 0.05 = 0.0375 ⇒ k_3 = intercept/slope = **0.10 h^{-1}**

The fitted line already runs to D = 1: it is used again in (b).

(b) Wash-out: iterate D = 1.0 − 0.3 $X_A(1 - X_A)$

Step	D in	X_A = (D + 0.1)/2.7	Loss term	D out
1	1.000	0.407	0.0724	0.928
2	0.928	0.381	0.0707	0.929
3	0.929	0.381	0.0708	**0.929**

D_{max} = **0.93 h^{-1}** with X_A = 0.38

(c) Residual glucose near D_{max}

D (h^{-1})	X_A	f	s (mg L^{-1})
0.5 (Problem 1)	0.222	0.552	24.6
0.8	0.333	0.867	130
0.9	0.370	0.970	646

A 12 % rise in D (0.8 → 0.9) multiplies s by 5: near D_{max} the culture is very sensitive and glucose leaves unused.

(d) Threshold at D = 0

X_A = 0.1/2.7 = 0.037; f = 0.3 × 0.037 × 0.963/1.0 = 0.0107; s_{min} = 0.020 × 0.0107/0.989 = **0.22 mg L^{-1}** — 55× below LAB

Loss term = $(1 - \gamma_{22})k_2X_A(1 - X_A)$ = 0.3 $X_A(1 - X_A)$. $D_{max} < \gamma_{11}k_1$ = 1.0 h^{-1} because ~7 % of the A made is lost when A is converted into G.

Solution 5 explained — the line and the threshold

The straight line of Problem 1 is a measuring tool; the threshold comes from the A → G loss

(a) What the line tells you

$$X_A = \frac{1}{\gamma_{22} k_2} D + \frac{k_3}{\gamma_{22} k_2}$$

slope = $1/(\gamma_{22}k_2)$ *intercept = $k_3/(\gamma_{22}k_2)$*

1. Slope gives the product $\gamma_{22}k_2 = 1/0.369 = 2.71\ h^{-1}$. $\gamma_{22} = 0.9$ is known, so $k_2 = 3.0\ h^{-1}$.
2. Intercept/slope cancels $\gamma_{22}k_2$: $k_3 = 0.0375/0.369 = 0.10\ h^{-1}$. No other parameter needed.
3. A non-zero intercept is the stand-by pool of Problem 1(e): it is how k_3 shows up in data.

(b) Why iteration works

With f = 1, μ = D gives one equation in D, but X_A itself depends on D:

$$D = \gamma_{11} k_1 - (1 - \gamma_{22}) k_2 X_A (1 - X_A) \;=\; 1.0 - 0.3\, X_A (1 - X_A)$$

The loss term is small and barely changes with D, so a guess of D = 1.0 is already within 8 % and two steps settle it. (Expanding gives a quadratic in D — same answer.)

(d) Where a threshold comes from

At D = 0 glucose uptake must still pay for the A lost in the A → G conversion: $\gamma_{11}k_1 f = (1 - \gamma_{22})k_2 X_A X_G$.

Williams: only the k_3 turnover keeps $X_A = 0.037$, so the loss is tiny → $s_{min} = 0.22$ mg L^{-1}. With $k_3 = 0$, $X_A = 0$ at D = 0 and $s_{min} = 0$, as in Monod.

LAB: the drain $v_2 = k_2 f_2 X_A$ saturates at lower s than synthesis, so it is large → $s_{min} = 12.3$ mg L^{-1}.

Both thresholds are model consequences of the A ⇄ G bookkeeping — neither model has an explicit maintenance term.

Problem 6 — LAB model: where is the best productivity?

Same Lactococcus culture as Problem 2, now fed with s_f = 2.0 g L^{-1} glucose

Given

Symbol	Value	Units
γ_{11}	0.50	g X_A (g S)$^{-1}$
γ_{22}	0.90	g X_G (g X_A)$^{-1}$
k_1 ; k_2	3.0 ; 0.95	h^{-1}
$K_{s,1}$; $K_{s,2}$	15 ; 5	mg L^{-1}
Feed s_f	**2.0**	**g L^{-1}**

Model

$$f_i = \frac{s}{s + K_{s,i}} \qquad \text{(results of Problem 2)}$$

$$D = \gamma_{11} k_1 f_1 - k_2 f_2$$

$$Y_{sx} = \gamma_{11} - (1 - \gamma_{22}) \frac{k_2}{k_1} \frac{f_2}{f_1}$$

Tasks

(a) From the biomass and glucose balances, show that the steady-state biomass is $x = Y_{sx}(s_f - s)$.

(b) Choose s = 50, 100, 200, 300, 500 and 1000 mg L^{-1}. For each, compute D, Y_{sx}, x and the productivity $P_x = Dx$.

(c) At which D is P_x highest? Express it as a fraction of D_{max} = 0.55 h^{-1}.

(d) A second plant has a dilute feed, s_f = 10 mg L^{-1}. What happens at any D?

(e) Why would you not run exactly at the optimum in practice?

Check your work: at s = 200 mg L^{-1}, x = 0.84 g L^{-1} and P_x = 0.39 g L^{-1} h^{-1}.

Builds on Problem 2 (VNL Ex. 7.6); parameters illustrative

Solution 6 — LAB model productivity

Productivity peaks close to, but safely below, the wash-out rate

(b) Steady states for $s_f = 2.0$ g L^{-1}

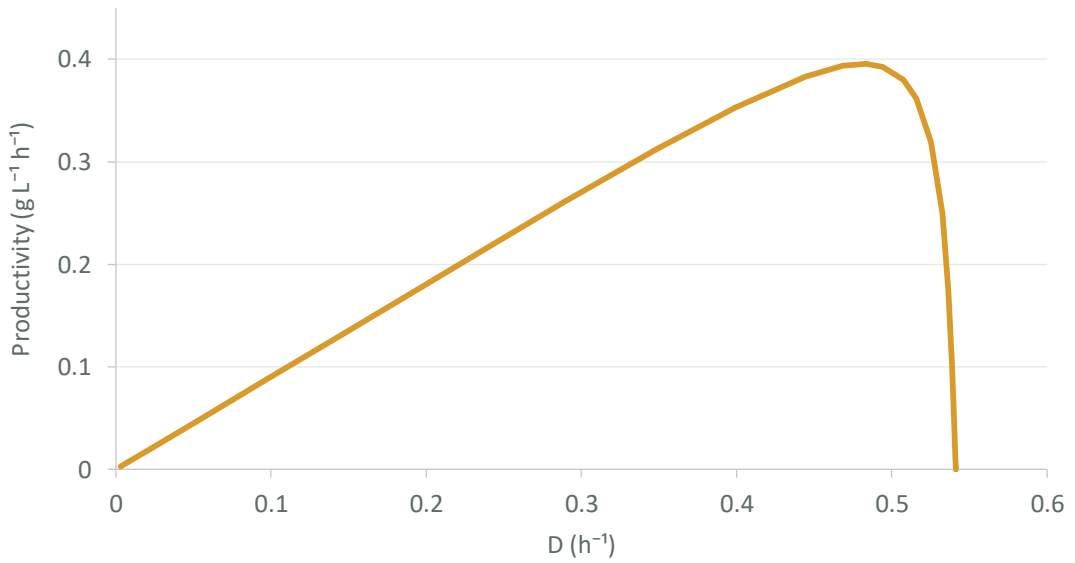
s (mg L^{-1})	D (h^{-1})	Y_{sx}	x (g L^{-1})	P_x
50	0.290	0.463	0.902	0.262
100	0.400	0.465	0.884	0.353
200	**0.469**	**0.467**	**0.840**	**0.394**
300	**0.494**	**0.467**	**0.794**	**0.393**
500	0.516	0.468	0.702	0.362
1000	0.533	0.468	0.468	0.249

(a) $x = Y_{sx}(s_f - s)$: see the explained slide.

(c) Optimum **$D \approx 0.48$ h^{-1}** ($s \approx 240$ mg L^{-1}), $x \approx 0.82$ g L^{-1}, **$P_x \approx 0.40$ g L^{-1} h^{-1}** — about **87 % of D_{max}**.

(d) $s_f = 10$ mg $L^{-1} < s_{min} = 12.3$ mg L^{-1}: even with no glucose used, $D = \gamma_{11}k_1f_1 - k_2f_2 < 0$. No steady state — the culture washes out at every D.

Productivity against dilution rate



(e) The curve is flat at the top ($P_x \geq 0.39$ for D = 0.46–0.50) but falls steeply above it: at D = 0.53 P_x is down to 0.25 and s has jumped to 1 g L^{-1}. A small upward drift in the pump costs a lot, a small downward drift almost nothing — so run a little below the optimum.

P_x in g L^{-1} h^{-1}. Y_{sx} hardly changes (0.463–0.468), so the trade-off is almost entirely about how much glucose is left unused.

Solution 6 explained — two balances and one trade-off

Biomass sets the growth rate, glucose sets how much biomass there is

(a) Deriving x, step by step

1. Biomass balance, sterile feed: growth equals wash-out.

$$\frac{dx}{dt} = (\mu - D)\,x = 0 \;\Rightarrow\; \mu = D$$

2. Glucose balance: feed in, outflow, uptake by cells (v_1 per g DW).

$$\frac{ds}{dt} = D(s_f - s) - v_1 x = 0 \;\Rightarrow\; x = \frac{D\,(s_f - s)}{v_1}$$

3. Yield is μ/v_1, and $\mu = D$:

$$Y_{sx} = \frac{\mu}{v_1} = \frac{D}{v_1} \;\Rightarrow\; x = Y_{sx}(s_f - s)$$

4. Productivity is biomass leaving per litre per hour:

$$P_x = D\,x = D\,Y_{sx}\,(s_f - s)$$

(b) Worked example: s = 200 mg L^{-1}

Quantity	Calculation	Result
f_1 ; f_2	200/215 ; 200/205	0.930 ; 0.976
D	1.5 × 0.930 − 0.95 × 0.976	0.469 h^{-1}
Y_{sx}	0.5 − 0.1 × 0.317 × 1.049	0.467
x	0.467 × (2.0 − 0.2)	0.840 g L^{-1}
P_x	0.469 × 0.840	0.394

The trade-off

Raising D processes more feed per hour (factor D) …

… but **s rises**, so less of the feed is turned into cells (factor $s_f - s$).

At low D the first effect wins, near D_{max} the second wins — so there is a maximum in between.

Note the units: s is in mg L^{-1} in f, but $s_f - s$ must be in g L^{-1} for x.

0.317 = k_2/k_1 = 0.95/3.0; 1.049 = f_2/f_1. Same yield formula as Solution 2.

KEY IDEAS

Before you start: the cybernetic toolkit

Problems 7–11 all use the same four moves, plus three shortcuts

The four moves

1 Saturation

one number per sugar, 0 to 1

$$f_i = \frac{s_i}{s_i + K_i}$$

2 Return

what pathway i can deliver now, with the enzyme it already has

$$r_i^* = k_i \, f_i \, X_{E,i}$$

3 Control

u: which enzyme to make; w: which enzyme to use

$$u_i = \frac{r_i^*}{\sum_j r_j^*}, \qquad w_i = \frac{r_i^*}{\max_j r_j^*}$$

4 Outcomes

growth rate, then each enzyme balance

$$\mu = \sum_i \gamma_i r_i^* \, w_i$$

Enzyme balance: synthesis $k_E f_i u_i$ – decay $k_{deg} X_{E,i}$ + constitutive k_{con} – dilution $\mu X_{E,i}$

Three shortcuts

A Only one sugar left

Then u = w = 1 for it: ordinary single-substrate growth (4e, 9, 10).

B Conditions steady

Enzyme at quasi-steady state: $X_E = (k_E f u + k_{con})/(k_{deg} + \mu)$ (8a, 9a, 10).

C Enzyme still small

Drop the small terms and solve a linear equation: $X_E \approx X_0 + At$ (4e, 9d), or exponential when $u \propto X_E$ (8).

Memory time ≈ $1/(k_{deg} + \mu)$

≈ 1.5–2 h when growing fast. Anything that changes faster than this, the enzyme levels still 'remember' the past.

Common parameters (Problem 4): glucose k = 2.0, K = 0.01 g L^{-1}, γ = 0.50; xylose k = 1.0, K = 0.05 g L^{-1}, γ = 0.45; k_E = 0.5, k_{deg} = 0.05, k_{con} = 0.001 h^{-1}.

Problem 7 — Cybernetic model: two enzymes, three sugars

Cells pre-grown on a sugar mixture arrive with both enzymes fully induced. Who gets the resources now?

Given

Parameter	Glucose	Xylose	Arabinose
k_i (h^{-1}, per unit enzyme)	2.0	1.0	1.5
K_i (g L^{-1})	0.01	0.05	0.10
γ_i (g DW g^{-1})	0.50	0.45	0.45
State: s_i (g L^{-1})	0.5	1.0	2.0
State: $X_{E,i}$	1.0	1.0	0.5

Model

$$r_i^* = k_i \frac{s_i}{s_i + K_i} X_{E,i}, \qquad u_i = \frac{r_i^*}{\sum_j r_j^*}, \qquad w_i = \frac{r_i^*}{\max_j r_j^*}$$

$$\mu = \sum_i \gamma_i r_i^* w_i$$

Tasks

(a) Glucose and xylose only (ignore arabinose): compute r_1^*, r_2^*, u_1, u_2, w_1, w_2. Is xylose being used?

(b) Compute μ three ways: with the cybernetic w; with no inhibition ($w_1 = w_2 = 1$); with a perfect switch (w = 1 for the best sugar, 0 for the other). Does the cybernetic rule give the fastest growth right now?

(c) With $X_{E,1} = X_{E,2} = 1$, below what glucose concentration does xylose become the preferred sugar? Compare with Problem 4(d).

(d) Now add arabinose. Compute r_3^* and the three u's and w's. Check that $\Sigma u_i = 1$.

(e) Glucose runs out, enzymes unchanged. Which sugar now has w = 1, and how strongly is the other used?

Check your work: $r_1^* = 1.96$ and $r_2^* = 0.952$ g S (g DW h)$^{-1}$.

Glucose and xylose parameters as Problem 4; arabinose values illustrative. Enzyme levels relative (fully induced ≈ 1).

Solution 7 — fully induced cells

With both enzymes present the model predicts simultaneous use — for now

(a), (d), (e) Control variables

Quantity	Glucose	Xylose	Arabinose
f_i	0.980	0.952	0.952
r_i^*	1.961	0.952	0.714
u_i (2 sugars)	0.673	0.327	–
w_i (2 sugars)	1	0.486	–
u_i (3 sugars)	0.541	0.263	0.197
w_i (3 sugars)	1	0.486	0.364
u_i (glucose gone)	0	0.571	0.429
w_i (glucose gone)	0	**1**	**0.75**

(a) Xylose uptake $r_2^* w_2 = 0.952 \times 0.486 =$ **0.463** — about a quarter of the glucose uptake. **Both sugars are used.**

(d) $\Sigma u = 0.541 + 0.263 + 0.197 = 1$ ✓; $\mu = 1.31\ h^{-1}$.

(e) Xylose takes the lead ($w = 1$); arabinose still runs at 75 %: no second diauxie between two similar sugars.

(b) Three control rules, same cells

Rule	w_1 ; w_2	μ (h^{-1})
Cybernetic	1 ; 0.486	**1.19**
No inhibition	1 ; 1	1.41
Perfect switch	1 ; 0	0.98

No: letting both enzymes run fully would grow faster. The cybernetic w is a description of catabolite inhibition, not an instantaneous optimum; the 'optimal' part of the model is u, the long-run return on enzyme synthesis.

(c) Switch point with equal enzyme

$$2.0\,\frac{s_1}{s_1 + 0.01} = 0.952 \;\Rightarrow\; s_1 = 9.1 \times 10^{-3}\ \mathrm{g\,L^{-1}}$$

9.1 mg L^{-1}, about 75× higher than in Problem 4(d) (0.12 mg L^{-1}), because xylose now has 50× more enzyme. The switch point is set by enzyme levels, not by the K_i alone.

r* in g S (g DW h)$^{-1}$. These μ values are high because both pathways are fully induced; this state does not last (next slide).

Solution 7 explained — simultaneous now, sequential later

The matching law slowly starves the poorer pathway, even when it starts fully induced

(a), (b) Line by line

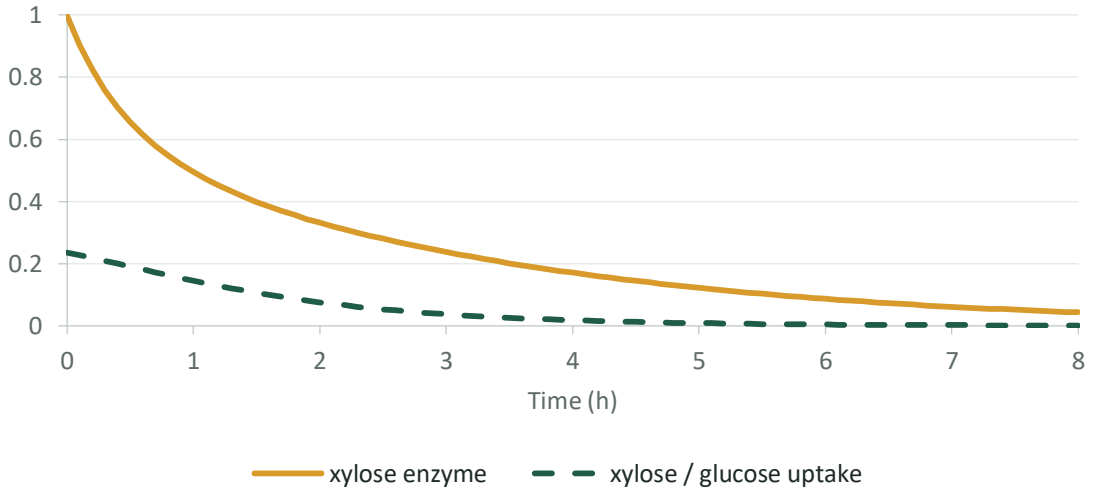
Quantity	Glucose (i = 1)	Xylose (i = 2)
$f_i = s_i/(s_i + K_i)$	0.5/0.51 = 0.980	1.0/1.05 = 0.952
$r_i^* = k_i f_i X_{E,i}$	2.0 × 0.980 × 1 = 1.961	1.0 × 0.952 × 1 = 0.952
$u_i = r_i^*/\Sigma r^*$	1.961/2.913 = 0.673	0.952/2.913 = 0.327
$w_i = r_i^*/\max r^*$	1	0.952/1.961 = 0.486
Uptake $r_i^* w_i$	1.961	0.463
$\gamma_i r_i^* w_i$	0.980	0.208

$\mu = 0.980 + 0.208 =$ **$1.19\ h^{-1}$**

Xylose is hit only *once* here: r_2^* is large because $X_{E,2} = 1$, so w_2 is about one half. In Problem 4 the same xylose had $X_{E,2} = 0.02$ and was hit twice (small r^*, then small w).

u_i and w_i always use the same r^*s: u divides by the sum, w by the largest.

What happens next (sugars held constant)



$u_2 = 0.33$ cannot keep up with dilution at $\mu \approx 1.2\ h^{-1}$, so $X_{E,2}$ falls. A smaller $X_{E,2}$ lowers r_2^*, which lowers u_2 and w_2 further: a self-reinforcing loop.

After ~5 h xylose uptake is below 1 % of glucose uptake: the culture has 'learned' to ignore xylose.

Exact curves: numerical integration of the full model from the state of Problem 7, with $s_1 = 0.5$ and $s_2 = 1.0$ g L^{-1} held constant (e.g. sugars topped up).

Problem 8 — Cybernetic model: a late glucose pulse

The reverse of Problem 4: now the cells carry the enzyme for the poorer sugar

Given

Parameter	Glucose	Xylose
k_i (h^{-1}) ; K_i (g L^{-1})	2.0 ; 0.01	1.0 ; 0.05
γ_i (g DW g^{-1})	0.50	0.45
$k_{E,i}$; $k_{deg,i}$ (h^{-1})	0.5 ; 0.05	0.5 ; 0.05
$k_{con,i}$ (h^{-1})	0.001	0.001
State: s_i (g L^{-1}) after pulse	1.0	1.0
State: $X_{E,i}$ before pulse	(a)	0.998

Before the pulse: balanced growth on xylose alone, $\mu = 0.428\ h^{-1}$ (Problem 4e). Assume s_1, s_2 stay ≈ constant afterwards.

Model

$$\frac{dX_{E,i}}{dt} = k_{E,i}\frac{s_i}{s_i + K_i}u_i - k_{\mathrm{deg},i}X_{E,i} + k_{\mathrm{con},i} - \mu X_{E,i}$$

$$\frac{dX}{dt} = \lambda X + c \quad \Rightarrow \quad X(t) = \left(X_0 + \frac{c}{\lambda}\right)e^{\lambda t} - \frac{c}{\lambda}$$

Tasks

(a) Before the pulse $u_1 = 0$. Show that $X_{E,1} = k_{con,1}/(k_{deg,1} + \mu)$ and evaluate it.

(b) Just after the pulse, compute r_1^*, r_2^*, u_1, w_1 and μ. Which sugar is being used?

(c) While $X_{E,1}$ is small, $u_1 \approx r_1^*/r_2^*$. Show that $dX_{E,1}/dt \approx \lambda X_{E,1} + k_{con,1}$ with $\lambda = k_{E,1}k_1f_1^2/r_2^* - (k_{deg,1} + \mu)$, and evaluate λ.

(d) Estimate the time until glucose becomes the preferred sugar ($r_1^* = r_2^*$), taking $X_{E,2} \approx 1$.

(e) Repeat (a), (b) and (d) with $k_{con,1} = 0.05\ h^{-1}$ (glucose uptake partly constitutive). What does k_{con} control?

Check your work: $X_{E,1}(0) = 0.0021$ and $\lambda \approx 0.55\ h^{-1}$.

Parameters as Problem 4 (VNL Sect. 7.4.2; Kompala et al. 1986). Enzyme levels are relative (fully induced ≈ 1).

Solution 8 — glucose pulse

The better sugar is ignored for hours unless the cell already has some of its enzyme

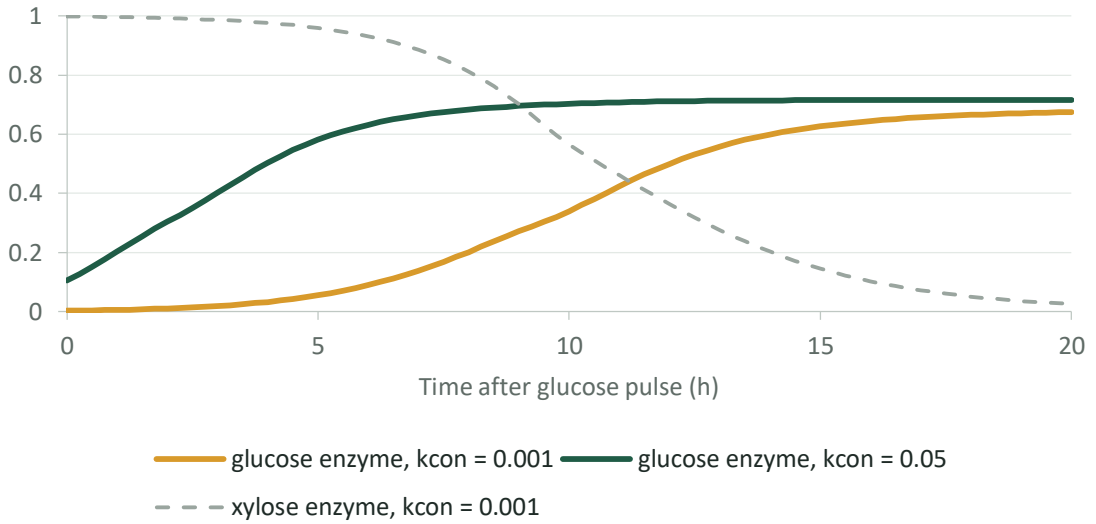
(a), (b), (d), (e) Results

Quantity	$k_{con,1} = 0.001$	$k_{con,1} = 0.05$
$X_{E,1}(0) = k_{con}/(0.05 + 0.428)$	0.0021	0.105
$r_1^* = 2 \times 0.990 \times X_{E,1}$	0.0041	0.207
$r_2^* = 0.952 \times 0.998$	0.951	0.951
u_1 ; w_1	0.0043 ; 0.0044	0.179 ; 0.218
μ just after pulse (h^{-1})	0.428	0.450
λ (h^{-1})	0.553	0.530
Time to $r_1^* = r_2^*$: estimate	**8.7 h**	**2.0 h**
Exact (simulation)	9.5 h	2.2 h

(b) Glucose uptake $r_1^* w_1 = 1.8 \times 10^{-5}$: the cells keep eating xylose; μ is unchanged.

(e) k_{con} sets the starting enzyme level, and the switch time depends on its logarithm: 50× more k_{con} → ~4× faster switch.

Glucose enzyme after the pulse (exact)



(d) Glucose wins when $X_{E,1} = r_2^*/(k_1 f_1) = 0.951/1.98 = 0.48$:

$t = \ln[(0.48 + 0.0018)/(0.0021 + 0.0018)]/0.553 = \ln(124)/0.553 =$ **8.7 h**

The estimate is a little short because $u_1 \approx r_1^*/r_2^*$ overstates u_1 once $X_{E,1}$ is no longer small.

$c/\lambda = k_{con}/\lambda = 0.0018$ ($k_{con} = 0.001$) or 0.094 ($k_{con} = 0.05$). Exact curves by numerical integration with $s_1 = s_2 = 1$ g L^{-1} held constant.

Solution 8 explained — the rich get richer

Enzyme for glucose is made in proportion to how much enzyme for glucose there already is

(c) Deriving λ, step by step

1. Enzyme balance for glucose (i = 1):

$$\frac{dX_{E,1}}{dt} = k_E f_1 u_1 + k_{con} - (k_{deg} + \mu) X_{E,1}$$

2. While $r_1^* \ll r_2^*$, the matching law is nearly a ratio:

$$u_1 = \frac{r_1^*}{r_1^* + r_2^*} \approx \frac{k_1 f_1 X_{E,1}}{r_2^*}$$

3. Substitute — $X_{E,1}$ appears in the synthesis term:

$$\frac{dX_{E,1}}{dt} \approx \left[\frac{k_E k_1 f_1^2}{r_2^*} - (k_{deg} + \mu)\right] X_{E,1} + k_{con}$$

4. Numbers:

$$\lambda = \frac{0.5 \times 2.0 \times 0.990^2}{0.951} - 0.478 = 1.031 - 0.478 = 0.553\ \mathrm{h}^{-1}$$

Why exponential?

The more glucose enzyme the cell has, the bigger r_1^*, so the more it invests in glucose enzyme ($u_1 \propto X_{E,1}$). It is autocatalytic — the same 'logistic' idea as X_A in Problem 3.

But it starts from almost nothing (0.002), so it needs ln(124) ≈ 4.8 e-foldings at 0.55 h^{-1}: about 9 h.

Is that realistic?

No: E. coli growing on xylose switches to glucose within minutes, because glucose uptake (the PTS) is always present and glucose blocks xylose uptake directly.

In the model this needs a much larger k_{con} for glucose (part e), or a different control law. Lesson: the 'small' constants k_{con} and k_{deg} decide how the model responds to a better sugar.

Compare Problem 4(e): there the enzyme had to be built with u = 1 (linear growth, ~1 h). Here u_1 starts near 0, so build-up is exponential and much slower.

Problem 9 — Cybernetic model: the whole diauxic batch

Problems 4 and 8 put together: glucose phase, switch, lag and xylose phase — by hand

Given

Quantity	Value	Units
Biomass x_0	0.05	g L^{-1}
Glucose $s_{1,0}$; xylose $s_{2,0}$	2.0 ; 2.0	g L^{-1}
$X_{E,1}(0)$; $X_{E,2}(0)$	0.68 ; 0.02	relative
Other parameters	as Problem 4	

Model

$$X_E = \frac{k_E f u + k_{con}}{k_{deg} + \mu} \qquad \text{(quasi-steady enzyme)}$$

$$x = x_0 e^{\mu t}, \qquad s_{1,0} - s_1 = \frac{x - x_0}{\gamma_1}$$

$$\frac{dX_{E,2}}{dt} = A - BX_{E,2} - CX_{E,2}^2 \quad \text{(Problem 4e)}$$

Tasks

(a) On glucose $u_1 \approx w_1 \approx 1$ and $f_1 \approx 1$. Find the quasi-steady $X_{E,1}$ and μ_1 from $dX_{E,1}/dt = 0$ with $\mu = \gamma_1 k_1 X_{E,1}$.

(b) When does glucose run out (t_1), and how much biomass is there then?

(c) Estimate $X_{E,2}$ at t_1. On glucose $u_2 \approx 0$, so $X_{E,2}$ decays towards $k_{con}/(k_{deg} + \mu_1)$ with rate $k_{deg} + \mu_1$.

(d) Estimate the lag (from t_1 until $\mu = \mu_\infty/2$) as in Problem 4(e), with f_2 at $s_2 = 2.0$ g L^{-1}. Repeat for $X_{E,2}(t_1) = 0.02$ and 0.2. Does the leftover enzyme matter much?

(e) Final biomass, and roughly when does the xylose run out? Sketch ln x and μ against t.

Check your work: $\mu_1 = 0.683\ h^{-1}$ and $t_1 \approx 4.5$ h.

Parameters as Problem 4 (VNL Sect. 7.4.2; Kompala et al. 1986). Enzyme levels are relative (fully induced ≈ 1). Inoculum pre-grown on glucose.

Solution 9 — the whole diauxic batch

Four short calculations reproduce the simulated curve to within about half an hour

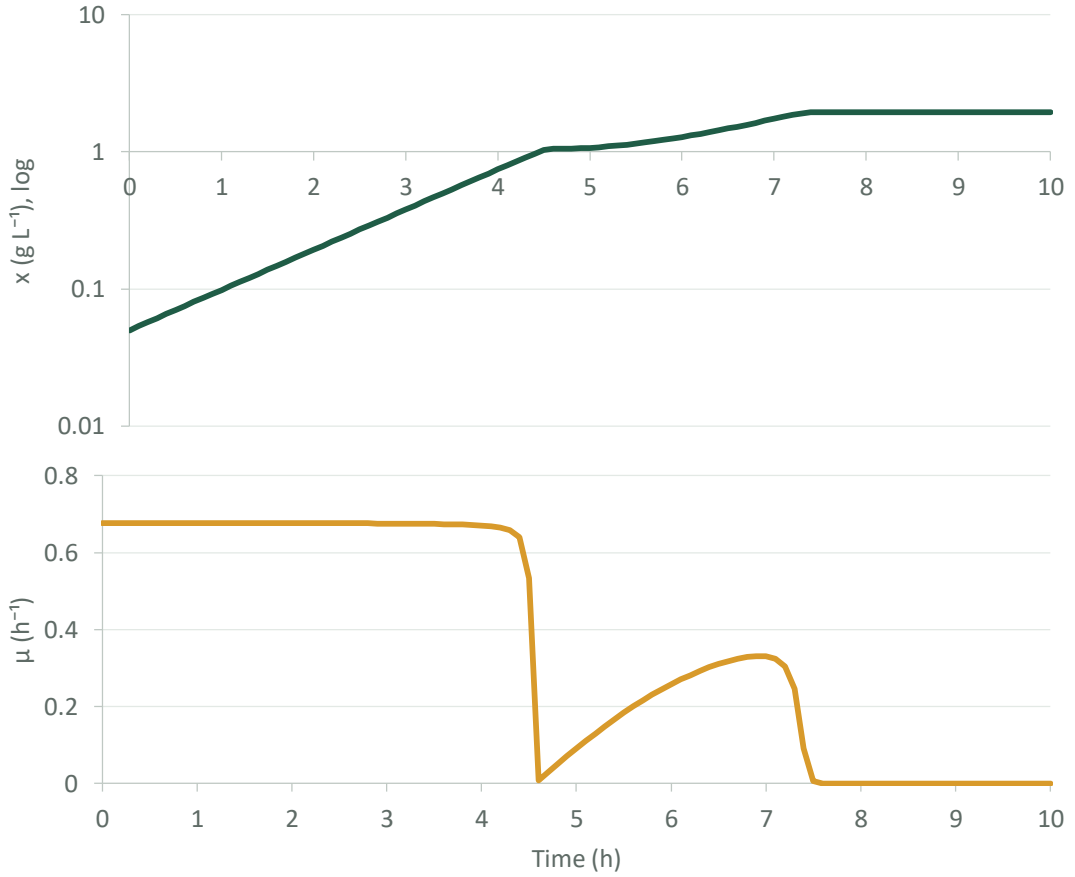
Estimates against the exact solution

Quantity	Hand estimate	Exact
(a) $X_{E,1} = \mu_1$ (h^{-1})	0.683	0.679
(b) t_1 ; $x(t_1)$	4.46 h ; 1.05 g L^{-1}	4.6 h ; 1.05 g L^{-1}
(c) $X_{E,2}(t_1)$	0.002	0.008
(d) Lag to $\mu_\infty/2$	1.0 h	1.1 h
(e) Xylose gone ; final x	≈ 6.9 h ; 1.95 g L^{-1}	7.4 h ; 1.95 g L^{-1}

(d) Does the leftover $X_{E,2}$ matter?

$X_{E,2}(t_1)$	0.002	0.02	0.2
Lag, estimate (h)	1.02	0.98	0.61
Lag, exact (h)	1.14	1.11	0.73

Hardly, unless a real stock is left (≥ 0.1). The lag is set mainly by how fast enzyme can be made (k_E), not by how little is left.



Exact curves by numerical integration of the full model (both sugars consumed). μ never reaches μ_∞ = 0.44 h^{-1}: the xylose runs out first.

Solution 9 explained — four phases, four short calculations

Each phase uses one shortcut from the toolkit slide

1 Glucose phase

0 – 4.5 h · shortcut B

$$X(0.05 + 1.0\,X) = 0.5 + 0.001$$

$$X^2 + 0.05X - 0.501 = 0$$

$$X_{E,1} = \mu_1 = 0.683\ \mathrm{h}^{-1}$$

$\gamma_1 k_1 = 1.0$, so μ_1 equals $X_{E,1}$.

2 Glucose runs out

t_1

$$x_1 = x_0 + \gamma_1 s_{1,0} = 1.05$$

$$t_1 = \frac{\ln(1.05/0.05)}{0.683}$$

$$t_1 = \frac{3.04}{0.683} = 4.46\ \mathrm{h}$$

Meanwhile $X_{E,2}$: 0.02 → 0.002 (decay rate 0.73 h^{-1}).

3 Lag

≈ 1 h · shortcut C

$$A = 0.5 \times 0.976 + 0.001 = 0.489$$

$$X_\infty \approx 1.0, \quad \mu_\infty = 0.44$$

$$t \approx \frac{0.50 - 0.002}{0.489} = 1.0\ \mathrm{h}$$

Half of μ_∞ needs $X_{E,2} = 0.50$.

4 Xylose phase

until ≈ 7 h · shortcut A

$$x_f = 1.05 + 0.45 \times 2.0 = 1.95$$

$$\Delta t \approx \frac{\ln(1.95/1.05)}{0.44} = 1.4\ \mathrm{h}$$

$$t_{end} \approx 4.5 + 1.0 + 1.4 = 6.9\ \mathrm{h}$$

Exact 7.4 h: μ only reaches 0.33 h^{-1}.

Two lessons. (1) The lag ≈ $(X_\infty/2)/k_E$ is set by how fast the cell can make the new enzyme; whether 0.02 or 0.002 is left hardly matters. Only a real stock (cells pre-grown on the mixture) shortens it.

(2) In a short xylose phase the culture never becomes fully adapted, so the simple 'jump to μ_∞' estimate ends too early.

$f_2 = 2.0/2.05 = 0.976$; $C = \gamma_2 k_2 f_2 = 0.439$; X_∞ from $CX^2 + 0.05X - A = 0$.

PROBLEM 10

Problem 10 — Cybernetic model in a chemostat on one sugar

Glucose-limited chemostat with sterile feed. With one sugar u = w = 1, so only the enzyme balance is left

Given

Symbol	Value	Units
k ; K	2.0 ; 0.01	h^{-1} ; g L^{-1}
γ	0.50	g DW (g S)$^{-1}$
k_E ; k_{deg} ; k_{con}	0.5 ; 0.05 ; 0.001	h^{-1}

Model

$$r^* = k\, f\, X_E, \qquad f = \frac{s}{s+K}, \qquad \mu = \gamma\, r^*$$

$$\frac{dX_E}{dt} = k_E f + k_{con} - k_{deg} X_E - \mu X_E$$

$$\text{Monod for comparison}: \quad \mu = D_{max}\frac{s}{s+K}$$

Tasks

(a) Show that at steady state $X_E = (k_E f + k_{con})/(k_{deg} + D)$ and $D = \gamma k f X_E$. Eliminate X_E to get $\gamma k k_E f^2 + \gamma k k_{con} f - D(k_{deg} + D) = 0$.

(b) Compute f, s and X_E at D = 0.1, 0.3 and 0.5 h^{-1}.

(c) Find D_{max} ($f \to 1$). Compare with μ_1 in Problem 9(a). Why are they the same?

(d) Compare s with a Monod model with the same K and $\mu_{max} = D_{max}$. How big are the differences, and where?

(e) Why does the model behave almost like Monod here but not in Problems 4, 8 and 9? Hint: compare $1/(k_{deg} + D)$ with how fast conditions change.

Check your work: at D = 0.3 h^{-1}, f = 0.457 and X_E = 0.656.

Glucose parameters of Problem 4. Single-substrate form of the Kompala model.

Solution 10 — cybernetic chemostat

At steady state the enzyme just follows D: the result is almost a Monod curve

(b), (d) Steady states

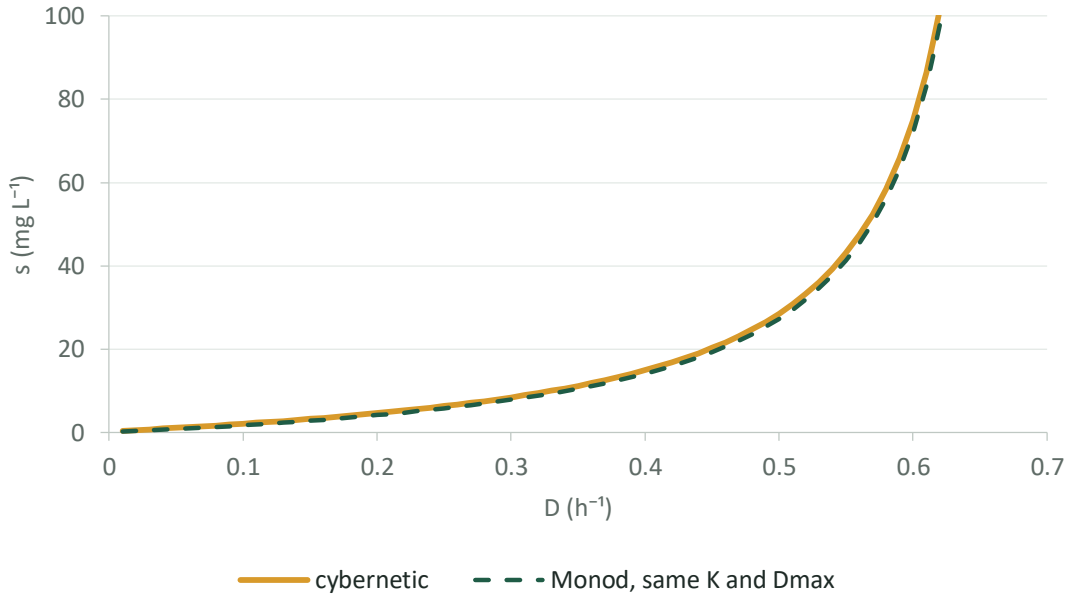
D (h^{-1})	f	s (mg L^{-1})	X_E	s Monod
0.10	0.172	2.1	0.581	1.7
0.30	0.457	8.4	0.656	7.8
0.50	0.741	28.6	0.675	27.3
0.65	0.953	202	0.682	195

(c) f = 1: $D^2 + 0.05D - 0.501 = 0 \Rightarrow$ **D_{max} = 0.683 h^{-1}**. It is the same equation as 9(a): at f = 1 the chemostat at wash-out and the batch on excess glucose are the same balanced growth.

(d) Within ~20 %, largest at low D, where X_E drops (0.58 at D = 0.1 vs 0.68 near D_{max}) so slightly more glucose is needed.

(e) At steady state X_E simply tracks D. The enzyme 'memory' $1/(k_{deg} + D)$ is 2–7 h; it matters only when conditions change faster than that, as in batch.

Residual glucose against dilution rate



The two curves almost coincide. With one sugar, the cybernetic model is essentially a Monod model with a slowly adjusting enzyme pool.

Monod: $s = KD/(D_{max} - D)$. Enzyme time constant $1/(k_{deg} + D)$: 6.7 h at D = 0.1, 1.8 h at D = 0.5 h^{-1}.

Solution 10 explained — algebra and a worked example

Same order of attack as Problems 1 and 2: internal balance first, then μ = D

(a) Deriving the quadratic

1. Biomass balance, sterile feed:

$$\mu = D$$

2. Enzyme balance at steady state (u = 1, μ = D):

$$0 = k_E f + k_{con} - (k_{deg} + D)X_E \;\Rightarrow\; X_E = \frac{k_E f + k_{con}}{k_{deg} + D}$$

3. Growth: $\mu = \gamma k f X_E = D$. Insert X_E:

$$D\,(k_{deg} + D) = \gamma k f\,(k_E f + k_{con})$$

4. Collect powers of f; take the positive root:

$$\underset{0.5}{\gamma k k_E}\, f^2 + \underset{0.001}{\gamma k k_{con}}\, f - D(k_{deg} + D) = 0$$

(b) Worked example: D = 0.3 h^{-1}

Quantity	Calculation	Result
c = D(k_{deg} + D)	0.3 × 0.35	0.105
f	[−0.001 + √(10^{-6} + 4 × 0.5 × 0.105)]/1.0	0.4573
s	0.01 × 0.4573/(1 − 0.4573)	8.42 mg L^{-1}
X_E	(0.5 × 0.4573 + 0.001)/0.35	0.656
Check μ	0.5 × 2.0 × 0.4573 × 0.656	0.300 ✓

Reading the result. Glucose enters only through f, and the enzyme only through X_E. Fixing D fixes both, so a cybernetic chemostat has no memory at steady state.

Compare Problem 2: there the threshold s_{min} came from a drain on A that saturates early. Here the enzyme drain $(k_{deg} + D)X_E$ is balanced by k_{con} even at s = 0, so there is no threshold.

Quadratic formula: f = [−b + √(b^2 + 4ac)]/(2a) with a = $\gamma k k_E$ = 0.5, b = $\gamma k k_{con}$ = 0.001.

Problem 11 — Cybernetic model: a mixed-sugar chemostat

Glucose and xylose fed together. Are they used together, or one after the other?

Given

Steady state	D = 0.1 h^{-1}	D = 0.5 h^{-1}
s_1 (mg L^{-1})	2.14	29.1
s_2 (mg L^{-1})	16.6	≈ 1000
$X_{E,1}$	0.301	0.672
$X_{E,2}$	0.421	0.010

Feed: $s_{f,1}$ = 1.0 g L^{-1} glucose + $s_{f,2}$ = 1.0 g L^{-1} xylose. Steady states found by simulation; other parameters as Problem 4.

Model

$$r_i^* = k_i \frac{s_i}{s_i + K_i} X_{E,i}, \qquad u_i = \frac{r_i^*}{\sum_j r_j^*}, \qquad w_i = \frac{r_i^*}{\max_j r_j^*}$$

$$\mu = \sum_i \gamma_i r_i^* w_i$$

$$x = \gamma_1 (s_{f,1} - s_1) + \gamma_2 (s_{f,2} - s_2)$$

Tasks

(a) For each D, compute f_1, f_2, r_1^*, r_2^*, u_1, u_2, w_1 and w_2.

(b) Check that $\mu = \Sigma \gamma_i r_i^* w_i$ equals D in both cases.

(c) Compute the biomass x at each D. Why is it roughly halved at D = 0.5 h^{-1}?

(d) Explain in words why both sugars are used at low D, but xylose leaves unused at D = 0.5 h^{-1}.

(e) The change between the two behaviours lies near D = 0.4 h^{-1}. Relate this to μ_∞ on xylose alone (Problem 4e: 0.428 h^{-1}).

Check your work: at D = 0.1 h^{-1}, $r_1^* \approx r_2^* \approx 0.105$ g S (g DW h)$^{-1}$.

Parameters as Problem 4 (VNL Sect. 7.4.2; Kompala et al. 1986). Enzyme levels are relative (fully induced ≈ 1).

Solution 11 — mixed feed

Low D: both sugars are used. High D: glucose takes everything and xylose washes through

(a) Control variables

	Glc, D = 0.1	Xyl, D = 0.1	Glc, D = 0.5	Xyl, D = 0.5
f_i	0.176	0.250	0.744	0.952
r_i^*	0.106	0.105	1.000	0.0095
u_i	0.502	0.498	0.991	0.0094
w_i	1	**0.993**	1	**0.0095**
$r_i^* w_i$	0.106	0.104	1.000	0.0001

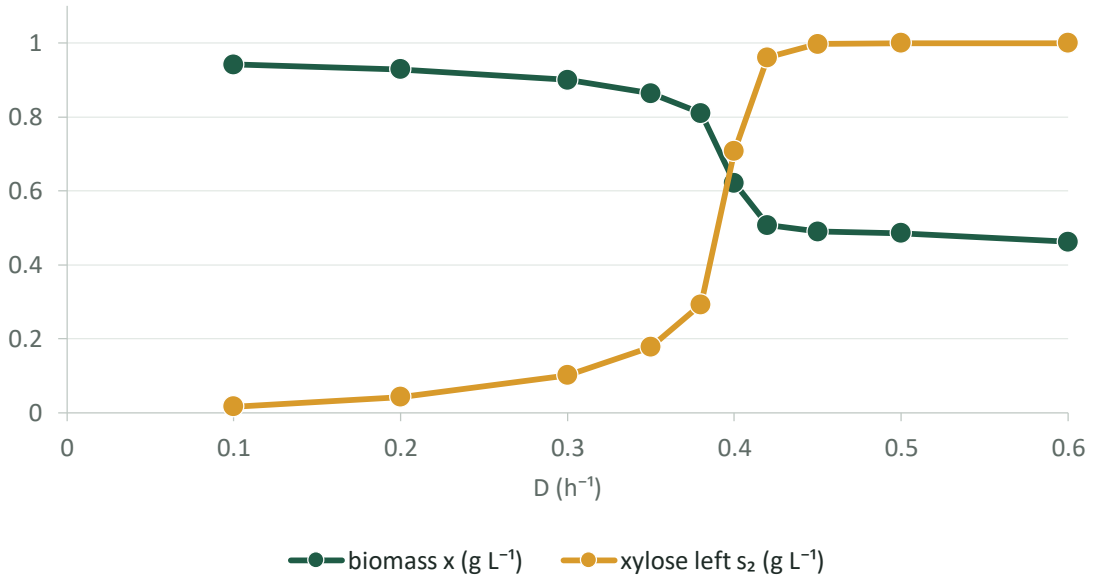
(b) D = 0.1: $0.5 \times 0.106 + 0.45 \times 0.104 = 0.053 + 0.047 =$ **0.100** ✓

D = 0.5: $0.5 \times 1.000 + 0.45 \times 0.0001 =$ **0.500** ✓

(c) x = **0.94** g L^{-1} at D = 0.1 but **0.49** at D = 0.5: at high D only the glucose half of the feed is turned into cells.

At D = 0.1 the two returns are almost equal, so the cell splits its investment 50 : 50.

Biomass and unused xylose against D (exact)



(e) The switch sits between D = 0.38 and 0.42 h^{-1} — just below μ_∞ = 0.43 h^{-1}, the fastest growth xylose alone can support.

Steady states by numerical integration of the chemostat form of the model (Problem 4 parameters; feed 1 + 1 g L^{-1}).

Solution 11 explained — where the switch comes from

Whether the xylose enzyme survives depends on whether its return can keep up with dilution

Low D: glucose is scarce

1. Glucose is pulled down to ~2 mg L^{-1}, so $f_1 \approx 0.18$ and r_1^* falls to the level of r_2^*.
2. The matching law then splits the investment about 50 : 50 ($u_1 \approx u_2 \approx 0.5$).
3. Growth is slow, so dilution μX_E is small: both enzymes are kept, $w_2 \approx 1$ and both sugars are used.

High D: glucose is plentiful

1. Fast growth needs glucose at ~30 mg L^{-1}, so $f_1 \approx 0.74$ and $r_1^* \approx 100 \times r_2^*$.
2. Almost all enzyme synthesis goes to glucose ($u_2 < 1$ %).
3. Dilution at 0.5 h^{-1} washes the xylose enzyme out of the cells faster than it is made: $X_{E,2} \rightarrow 0.01$.

(e) Why near 0.43 h^{-1}?

Even if the cell put all its enzyme-making into xylose ($u_2 = 1$), the xylose enzyme could only support $\mu_\infty = 0.428$ h^{-1} (Problem 4e).

Above that D, xylose cannot sustain its own enzyme against dilution. Because glucose also takes part of u, $X_{E,2}$ collapses slightly earlier.

D (h^{-1})	s_2 left (mg L^{-1})	$X_{E,2}$	w_2
0.30	101	0.470	0.95
0.38	293	0.458	0.85
0.40	707	0.369	0.55
0.42	961	0.171	0.20
0.45	997	0.050	0.05

Carbon-limited chemostats often show this: mixed sugars used together at low D, one after the other near the maximum growth rate.

Values from the same simulations as the solution slide.